UC SAN DIEGO / ECE 251C

12
Frequency decomposition and sparsity

Octave bands, time–frequency localization, and vanishing moments

Why can a signal with many nonzero samples have only a few nonzero wavelet coefficients?

12.1 Reading the DWT as a frequency decomposition

The discrete wavelet transform is an invertible change of coordinates. Invertibility alone, however, does not explain why those coordinates are useful. We now connect the filter-bank computation to two properties: a frequency decomposition into octaves, and cancellation of simple signal behavior away from transitions.

At the finest available resolution I, write

$$f_I(t) = \sum_n a_I[n]\phi_{I,n}(t), \qquad \phi_{i,n}(t) = 2^{i/2}\phi(2^i t - n).$$

Here $a_i[n] = \langle f, \phi_{i,n}\rangle_{L^2}$, and the wavelet coefficients are $d_i[n] = \langle f, \psi_{i,n}\rangle_{L^2}$, with $\psi_{i,n}(t) = 2^{i/2}\psi(2^i t - n)$. The function f_I is the projection of f onto V_I. One analysis stage computes

$$a_{i-1}[n] = (h_0 * a_i)[2n], \qquad d_{i-1}[n] = (h_1 * a_i)[2n]. \tag{12.1}$$

The low-pass output describes V_{i-1}; the detail output describes W_{i-1}, where $V_i = V_{i-1} \oplus W_{i-1}$. Only the approximation branch is decomposed again in the usual DWT.

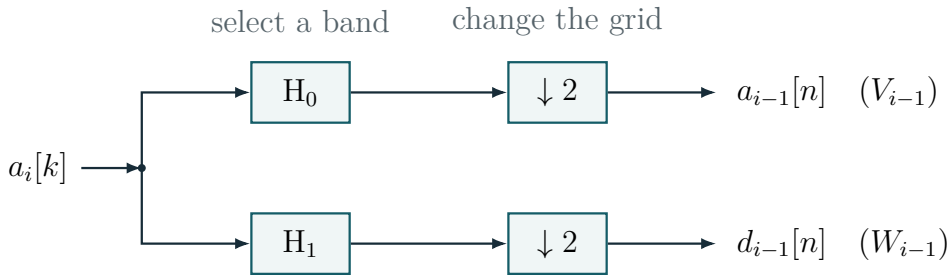


Figure 12.1: One analysis stage separates approximation and detail. Downsampling changes the frequency coordinates of both outputs.

12.2 What downsampling does to the frequency axis

The frequency of a discrete sequence is measured in radians *per sample of that sequence*. Downsampling changes the sample spacing, so the same physical oscillation receives a different numerical frequency label. Keeping track of that label is the first task.

Use the DTFT convention

$$\widehat{x}(\omega) = X(\mathrm{e}^{\mathrm{j}\omega}) = \sum_{n\in\mathbb{Z}} x[n]\mathrm{e}^{-\mathrm{j}\omega n}.$$

Suppose $y[n] = x[2n]$. We use θ for the output's frequency variable. Starting with finitely supported sequences makes every rearrangement finite:

$$\begin{aligned}
&\frac{1}{2}\left[\widehat{x}(\theta/2) + \widehat{x}(\theta/2+\pi)\right] \\
&\quad = \sum_{k} x[k]\mathrm{e}^{-\mathrm{j}\theta k/2}\frac{1+\mathrm{e}^{-\mathrm{j}\pi k}}{2} \\
&\quad = \sum_{k \text{ even}} x[k]\mathrm{e}^{-\mathrm{j}\theta k/2} \\
&\quad = \sum_{n} x[2n]\mathrm{e}^{-\mathrm{j}\theta n} = \widehat{y}(\theta).
\end{aligned}$$

The factor $(1+(-1)^k)/2$ is one on even indices and zero on odd indices. Thus

$$\boxed{\widehat{y}(\theta) = \frac{1}{2}\left[\widehat{x}(\theta/2) + \widehat{x}(\theta/2+\pi)\right].} \tag{12.2}$$

The second argument can also be written $(\theta - 2\pi)/2$: the two versions differ by one DTFT period. The identity extends to ℓ^2 sequences as an equality of L^2 frequency responses, by finite truncation and Plancherel.

Track one oscillation. If $x[n] = \mathrm{e}^{\mathrm{j}\omega n}$, then

$$y[n] = \mathrm{e}^{\mathrm{j}(2\omega)n}.$$

Hence the output label is $\theta = 2\omega$ modulo 2π. In particular,

$$\omega = \frac{\pi}{4} \;\mapsto\; \theta = \frac{\pi}{2}, \qquad \omega = \frac{3\pi}{4} \;\mapsto\; \theta = -\frac{\pi}{2}.$$

These exponentials are diagnostic inputs, not finite-energy sequences. The second example shows the frequency folding that occurs in the high-pass branch.

If filtering precedes downsampling, replace $\widehat{x}$ in (12.2) by $\widehat{h}\,\widehat{x}$. The filter selects a band *before* the frequency axis is expanded and wrapped.

12.3 An idealized first split

For a full-band input, consider ideal analysis magnitudes

$$|\widehat{h}_0(\omega)| = \sqrt{2}\,\mathbf{1}_{\{|\omega|<\pi/2\}}, \qquad |\widehat{h}_1(\omega)| = \sqrt{2}\,\mathbf{1}_{\{\pi/2<|\omega|<\pi\}},$$

extended periodically. Endpoint values do not affect L^2 statements. These are idealized bands, not FIR filters; phases can be chosen to obtain the Shannon orthogonal bank. We are tracking support here, not prescribing those phases.

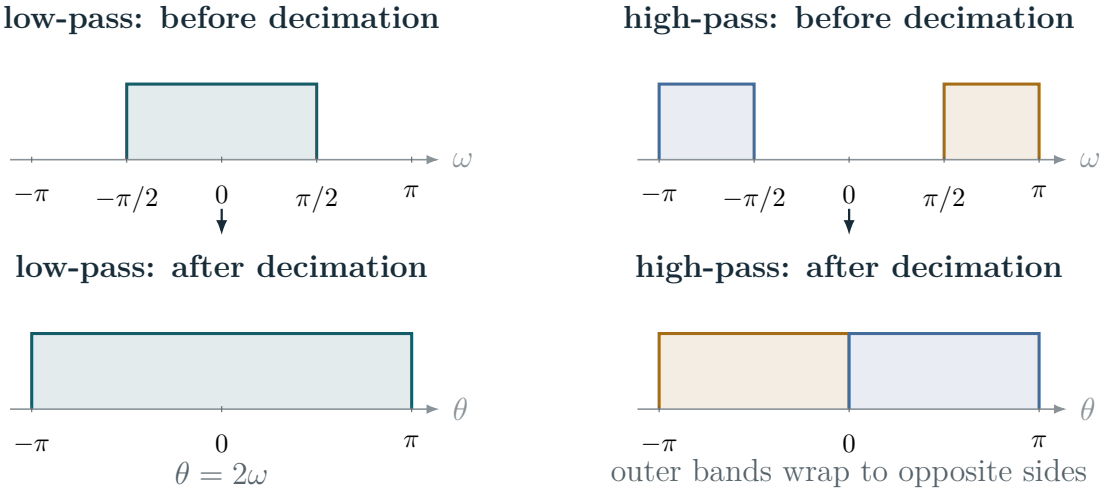


Figure 12.2: Ideal band locations before and after downsampling. Heights indicate occupied bands only, not amplitudes. The colors track which original high-frequency half-band goes to each output half.

In the low-pass branch, only the central half-band survives. For $-\pi < \theta < \pi$, the shifted term in (12.2) vanishes, so

$$\widehat{a}_{I-1}(\theta) = \tfrac{1}{2}\widehat{h}_0(\theta/2)\widehat{a}_I(\theta/2).$$

The surviving band $[-\pi/2, \pi/2]$ fills the new interval $[-\pi, \pi]$. Its amplitude is also multiplied by $1/2$ in this formula; the diagram does not suppress that factor mathematically.

The detail branch carries the two outer bands. Downsampling folds their positive and negative parts into its new principal interval. Its coefficients still represent the *original* high-frequency band, even though their output frequency interval is again $[-\pi, \pi]$. To interpret the tree, always label a branch by its frequencies on the original input grid.

12.4 Repeated low-pass splitting produces octaves

After one low-pass stage, the branch frequency is $\theta_1 = 2\omega$ on the retained band. After k low-pass stages it is $\theta_k = 2^k\omega$. The next low-pass filter again retains $|\theta_k| < \pi/2$, now meaning $|\omega| < \pi/2^{k+1}$.

Induction gives the bands below, all in the *original* variable ω:

$$\boxed{\begin{aligned} V_{I-k}: &\quad |\omega| \le \frac{\pi}{2^k}, \\ W_{I-k}: &\quad \frac{\pi}{2^k} \le |\omega| \le \frac{\pi}{2^{k-1}}, \qquad k \ge 1. \end{aligned}} \tag{12.3}$$

The space names label the reconstructed subspaces. These frequency identifications are exact for ideal Shannon bands and nominal for general wavelets.

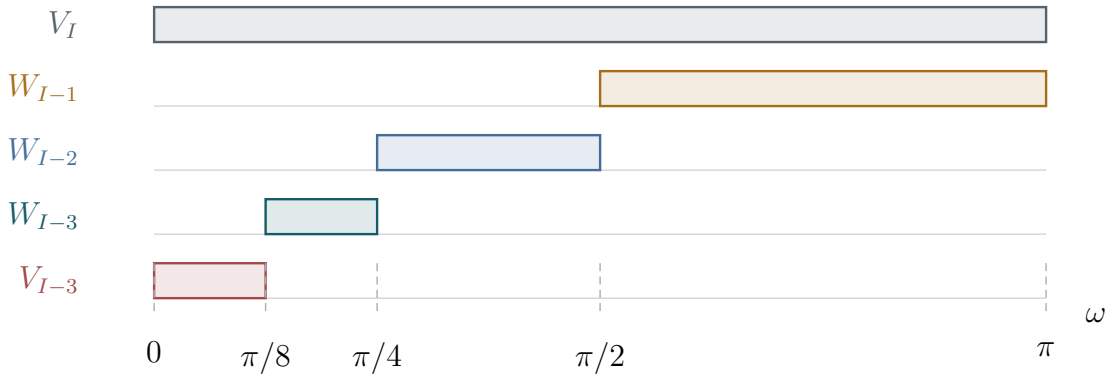


Figure 12.3: Three levels on one frequency axis, showing positive frequencies. The negative-frequency bands are their reflections. Bandwidth halves as the decomposition moves down the low-pass branch.

The orthogonal identity itself is exact for every orthogonal MRA:

$$V_I = V_{I-3} \oplus W_{I-3} \oplus W_{I-2} \oplus W_{I-1}.$$

Each positive detail band has upper endpoint twice its lower endpoint: it has width one on the $\log_2 \omega$ axis. This is the *octave* structure, named for the frequency ratio of two in a musical octave. It also motivates an analogy with auditory filter banks, not an identification of hearing with a dyadic DWT.

Discrete and continuous frequencies. If Ω denotes continuous-domain angular frequency, direct transformation of f_I gives

$$\widehat{f}_I(\Omega) = 2^{-I/2}\widehat{\phi}(2^{-I}\Omega)\widehat{a}_I(2^{-I}\Omega). \tag{12.4}$$

Thus $\omega = 2^{-I}\Omega$ on the finest coefficient grid. The extra factor $\widehat{\phi}$ matters: an arbitrary V_I is not a brick-wall bandlimited space.

12.5 Practical filters have overlapping transition bands

The sharp bands organize the DWT, but are not the exact spectra of compactly supported wavelets. Haar has normalized analysis powers

$$q_0(\omega) := \frac{|\hat{h}_0(\omega)|^2}{2} = \cos^2(\omega/2), \qquad q_1(\omega) := \frac{|\hat{h}_1(\omega)|^2}{2} = \sin^2(\omega/2).$$

They satisfy $q_0 + q_1 = 1$, but neither is a rectangular band.

Define the normalized three-level branch profiles by

$$A_3(\omega) = \prod_{r=0}^{2} q_0(2^r\omega),$$
$$D_k(\omega) = q_1(2^{k-1}\omega) \prod_{r=0}^{k-2} q_0(2^r\omega), \qquad k = 1, 2, 3.$$

An empty product equals one. Each deeper filter acts on a downsampled grid. The equivalent depth-k filter on the original grid is

$$B_k(z) = H_1(z^{2^{k-1}}) \prod_{r=0}^{k-2} H_0(z^{2^r}), \qquad D_k(\omega) = 2^{-k}|B_k(\mathrm{e}^{\mathrm{j}\omega})|^2.$$

For example, two filtering stages combine as $\sum_{m,r} h_0[m]h_0[r]x[4n-2m-r]$, giving $H_0(z^2)H_0(z)$ before decimation by four.

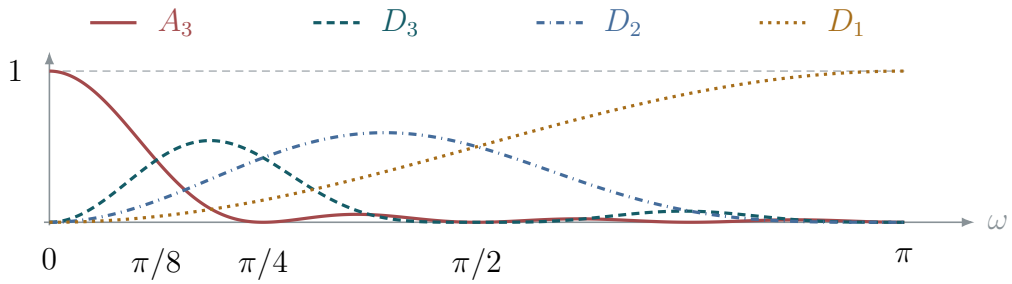


Figure 12.4: Actual normalized Haar branch powers. Transition regions overlap, and deeper bands have sidelobes. The four profiles sum to one, but are not disjoint frequency windows.

Put $A_k = \prod_{r=0}^{k-1} q_0(2^r\omega)$ and $A_0 = 1$. Then $D_k = A_{k-1} - A_k$, so

$$A_3 + D_3 + D_2 + D_1 = 1.$$

These are filter-power profiles, not pointwise output-energy formulas after decimation. Overlap is compatible with orthogonality.

12.6 Four ways to allocate time–frequency resolution

A coefficient is an inner product with an analyzing function q:

$$c_q = \langle f, q \rangle_{L^2} = \int_{\mathbb{R}} f(t) q^*(t)\, \mathrm{d}t = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\Omega) \hat{q}(\Omega)^*\, \mathrm{d}\Omega.$$

The first integral describes its time localization; the second describes its frequency localization. A rectangle records where q and $\hat{q}$ are concentrated. It is *not* a claim that a nonzero atom has compact support in both domains.

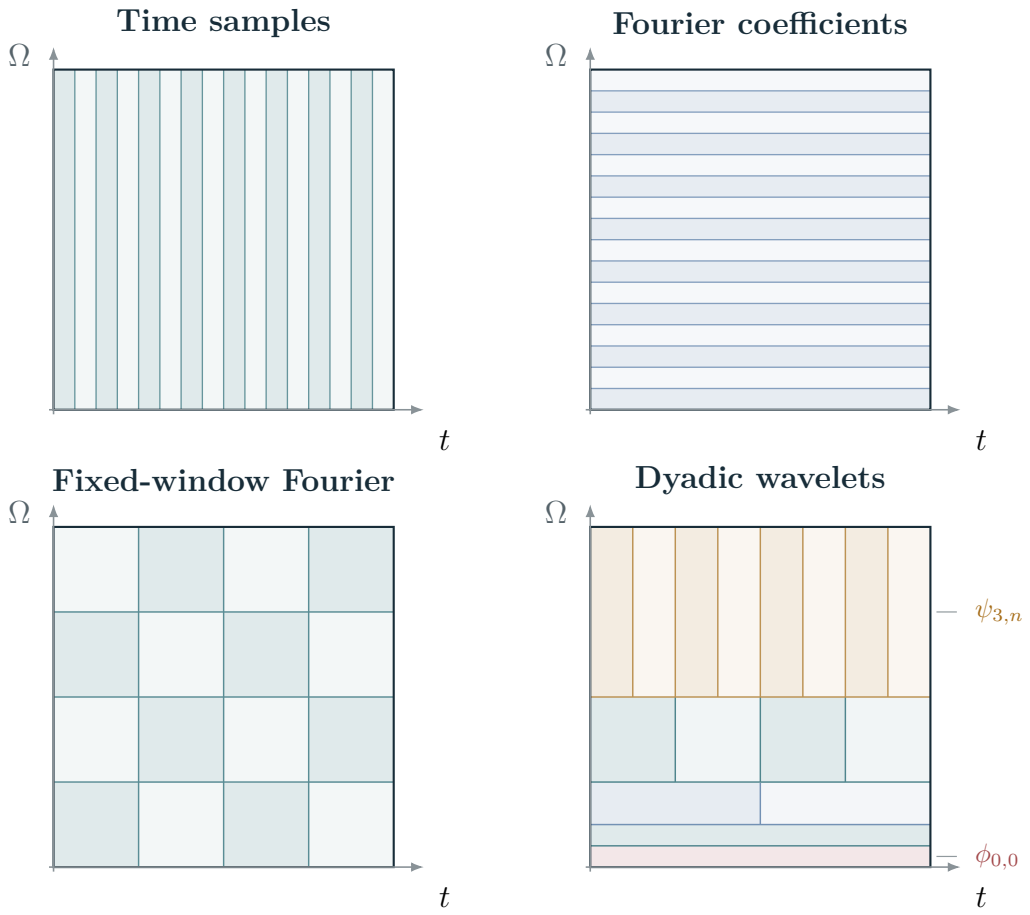


Figure 12.5: Each panel has 16 equal-area nominal coefficient cells, not exact joint supports. The wavelet panel retains one scaling cell, followed by 1, 2, 4, and 8 detail cells as frequency increases.

For an N-sample signal, the unit impulse $e_n[k] = \delta[k-n]$ has unitary DFT

$$(\mathrm{F}e_n)[m] = N^{-1/2} \mathrm{e}^{-\mathrm{j}2\pi mn/N}.$$

It occupies one time sample and all frequency bins. Conversely, the Fourier vector $N^{-1/2}\mathrm{e}^{\mathrm{j}2\pi mk/N}$ has constant magnitude over the whole record and occupies one DFT bin. These give the vertical and horizontal strips.

Fixed windows give fixed resolution. For a unit-norm window w, define

$$q_{u,\xi}(t) = w(t-u)\mathrm{e}^{\mathrm{j}\xi t}, \qquad V_w f(u,\xi) = \langle f, q_{u,\xi}\rangle_{L^2}.$$

This is the short-time Fourier transform (STFT); $|V_w f|^2$ is the spectrogram. Substituting $s = t - u$ gives

$$\widehat{q}_{u,\xi}(\Omega) = \mathrm{e}^{-\mathrm{j}(\Omega-\xi)u}\widehat{w}(\Omega-\xi).$$

Only the centers change. The time and frequency widths remain those of w and $\widehat{w}$. Taking more STFT samples does not shorten the window or narrow its spectrum.

Wavelet dilation changes both widths. For $\psi_{i,n}(t) = 2^{i/2}\psi(2^i t - n)$, put $s = 2^i t - n$:

$$\begin{aligned}\widehat{\psi}_{i,n}(\Omega) &= 2^{i/2}\int_{\mathbb{R}} \psi(2^i t - n)\mathrm{e}^{-\mathrm{j}\Omega t}\,\mathrm{d}t \\ &= 2^{-i/2}\int_{\mathbb{R}} \psi(s)\mathrm{e}^{-\mathrm{j}2^{-i}\Omega(s+n)}\,\mathrm{d}s \\ &= 2^{-i/2}\mathrm{e}^{-\mathrm{j}2^{-i}n\Omega}\widehat{\psi}(2^{-i}\Omega).\end{aligned}$$

An interval containing a fixed fraction of the time energy is compressed by 2^{-i}; the corresponding frequency region is expanded by 2^i. Thus

$$\boxed{\Delta t_i = 2^{-i}\Delta t_\psi, \quad \Delta\Omega_i = 2^i\Delta\Omega_\psi, \quad \Delta t_i\Delta\Omega_i = \Delta t_\psi\Delta\Omega_\psi.} \tag{12.5}$$

The translation step is also 2^{-i}: neighboring coefficients at a finer scale interrogate more closely spaced locations.

Nominal cells versus actual localization. A precise convention for the positive-frequency diagram is

$$[2^{-i}n, 2^{-i}(n+1)) \times [2^i\pi, 2^{i+1}\pi),$$

with a bottom scaling row $[n, n+1) \times [0, \pi)$. Every cell has area π. These cells partition coefficient locations; actual wavelets can extend beyond their cell. Real wavelets have symmetric positive- and negative-frequency energy, so the drawing shows only one side of their spectrum.

For a unit-norm atom q, suppose $\int_{\mathbb{R}\setminus T}|q(t)|^2\,\mathrm{d}t \le \varepsilon$. Then Cauchy–Schwarz gives

$$\left|\langle f, q\rangle - \int_T f(t)q^*(t)\,\mathrm{d}t\right| \le \sqrt{\varepsilon}\,\|f\|_{L^2}.$$

Thus little atom energy outside T limits the contribution from that region. Parseval gives the analogous statement for a frequency-concentration region.

12.7 What the uncertainty principle does and does not say

For a unit-norm atom q, suppose the following means and variances exist:

$$\tau = \int t|q(t)|^2\,\mathrm{d}t, \qquad \nu = \frac{1}{2\pi}\int \Omega|\widehat{q}(\Omega)|^2\,\mathrm{d}\Omega,$$

$$\sigma_t^2 = \int (t-\tau)^2|q(t)|^2\,\mathrm{d}t, \quad \sigma_\Omega^2 = \frac{1}{2\pi}\int (\Omega-\nu)^2|\widehat{q}(\Omega)|^2\,\mathrm{d}\Omega.$$

All integrals here are over $\mathbb{R}$. The uncertainty inequality is

$$\boxed{\sigma_t\sigma_\Omega \geq \tfrac{1}{2}.} \tag{12.6}$$

Proof for smooth, rapidly decaying atoms. Translation and modulation let us assume $\tau = \nu = 0$ without changing the variances. Integration by parts, followed by Cauchy–Schwarz, gives

$$\begin{aligned} 1 &= \int |q(t)|^2\,\mathrm{d}t = -\int t\frac{\mathrm{d}}{\mathrm{d}t}|q(t)|^2\,\mathrm{d}t \\ &= -2\,\mathrm{Re}\int tq(t)^*q'(t)\,\mathrm{d}t \leq 2\|tq\|_{L^2}\|q'\|_{L^2}. \end{aligned}$$

The boundary term is zero by decay. Plancherel and $\widehat{q'}(\Omega) = \mathrm{j}\Omega\widehat{q}(\Omega)$ give $\|tq\|_{L^2} = \sigma_t$ and $\|q'\|_{L^2} = \sigma_\Omega$, proving the bound. □

For general finite-variance atoms the same argument follows by smooth truncation. The inequality is a lower bound, not an assertion that all atoms have the same uncertainty product. The equal areas within one wavelet family come from reciprocal dilation in (12.5), not from equality in (12.6).

Haar requires a different notion of width. Direct integration yields

$$\widehat{\psi}(\Omega) = \frac{(1-\mathrm{e}^{-\mathrm{j}\Omega/2})^2}{\mathrm{j}\Omega}, \qquad \Omega^2|\widehat{\psi}(\Omega)|^2 = 16\sin^4(\Omega/4).$$

The first formula is understood at $\Omega = 0$ by continuity. The last function has infinite integral, so Haar has infinite frequency variance. Its finite-height boxes are nominal bands or energy-concentration regions, not finite-variance boxes. This does not invalidate their dyadic scaling law.

12.8 Haar coefficients vanish on constant pieces

Frequency localization explains the geometry of the representation. To understand sparsity, we need a cancellation property as well. Recall

$$\phi(t) = \mathbf{1}_{[0,1)}(t), \qquad \psi(t) = \mathbf{1}_{[0,1/2)}(t) - \mathbf{1}_{[1/2,1)}(t).$$

The Haar wavelet has zero mean:

$$\int_{\mathbb{R}} \psi(t)\,\mathrm{d}t = \frac{1}{2} - \frac{1}{2} = 0.$$

The positive and negative halves therefore cancel whenever the signal is constant over the wavelet's support.

Let $Q_{i,n} = [2^{-i}n, 2^{-i}(n+1))$, with left and right halves $Q_{i,n}^{\mathrm{L}}$ and $Q_{i,n}^{\mathrm{R}}$. Then

$$\boxed{d_i[n] = 2^{i/2}\left[\int_{Q_{i,n}^{\mathrm{L}}} f(t)\,\mathrm{d}t - \int_{Q_{i,n}^{\mathrm{R}}} f(t)\,\mathrm{d}t\right].} \tag{12.7}$$

If $f = c$ on $Q_{i,n}$, both integrals equal $c2^{-i-1}$, so $d_i[n] = 0$.

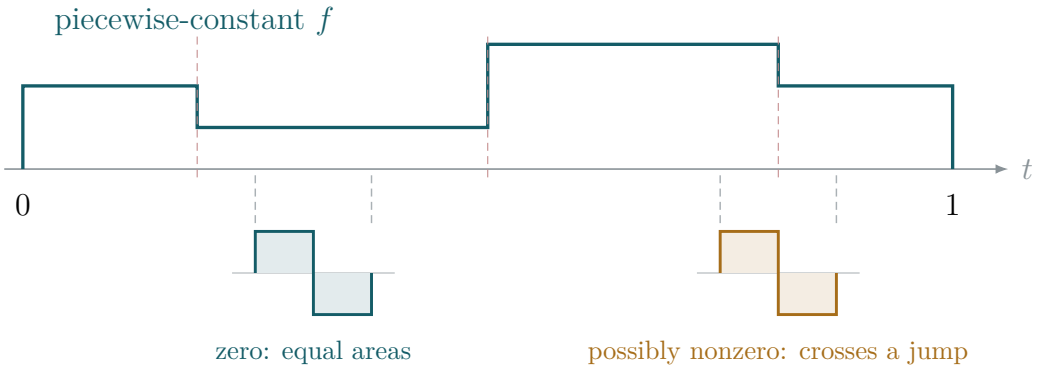


Figure 12.6: A Haar atom inside a flat piece has zero coefficient. Only supports that straddle transitions can contribute. The small atoms are drawn with a common illustrative height; their actual normalization is $2^{i/2}$.

For a piecewise-constant signal, a nonzero Haar detail can occur only when the *interior* of its supporting interval contains a jump. A jump at a cell boundary does not create a detail in either adjacent cell, since both cells are individually constant. A wavelet crossing several jumps can also have zero coefficient through cancellation; crossing a jump is necessary, not sufficient, for a nonzero coefficient.

This is a local statement, not an assertion that a discontinuous signal contains only low frequencies. A jump has Fourier content at arbitrarily high frequencies, but localized wavelets need to record it only near its location.

The coefficient of one jump. On $Q_{i,n} = [a, a+\ell)$, where $\ell = 2^{-i}$, suppose

$$f(t) = c + A\mathbf{1}_{[\tau,\infty)}(t), \qquad \tau = a + v\ell, \quad 0 < v < 1.$$

Here A is the signed jump size. The constant c contributes zero. If $0 < v \le 1/2$, the jump contributes on a length $(1/2 - v)\ell$ in the left half and $\ell/2$ in the right half. If $1/2 \le v < 1$, it contributes only on a length $(1 - v)\ell$ in the right half. Hence

$$d_i[n] = -A2^{-i/2} \begin{cases} v, & 0 \le v \le 1/2, \\ 1 - v, & 1/2 \le v \le 1. \end{cases} \tag{12.8}$$

In particular, $|d_i[n]| \le |A|2^{-i/2}/2$. The negative sign for an upward jump follows from our convention that the right half of ψ is negative.

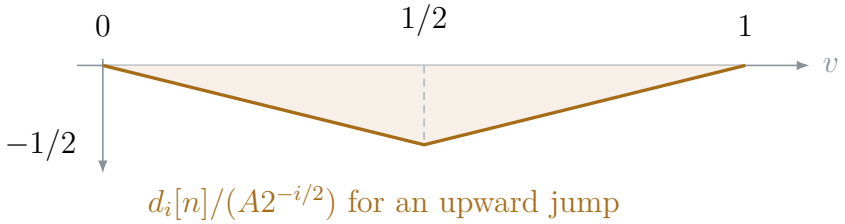


Figure 12.7: The coefficient depends on the jump location within the cell. It is largest in magnitude at the midpoint and vanishes at either boundary.

How many coefficients can be nonzero? Suppose f is piecewise constant on $[0, 1]$ with S interior jumps and is zero outside this interval. At scale $i \ge 0$, the dyadic cells partition $[0, 1]$. Each jump lies in at most one cell interior, so

$$\#\{n : d_i[n] \ne 0\} \le \min(S, 2^i).$$

At resolutions $0 \le i < I$, at most SI detail coefficients can be nonzero, out of $2^I - 1$ possible details.

The coarse Haar coefficient is just the average:

$$a_0[0] = \int_0^1 f(t)\,\mathrm{d}t, \qquad a_0[n] = 0 \quad (n \ne 0).$$

The full expansion is

$$f = a_0[0]\phi + \sum_{i=0}^{\infty} \sum_{n=0}^{2^i-1} d_i[n]\psi_{i,n} \quad \text{in } L^2.$$

A non-dyadic jump can persist at infinitely many scales: sparsity at each scale does not imply a finite expansion.

12.9 A concrete sparse expansion

To put numbers on the piecewise-constant picture, take

$$f(t) = 2\mathbf{1}_{[0,1)}(t) - \mathbf{1}_{[3/16,1/2)}(t) + \mathbf{1}_{[1/2,13/16)}(t).$$

Its successive heights on $[0,1]$ are $2, 1, 3, 2$, with jumps at $3/16$, $1/2$, and $13/16$; it is zero outside $[0,1]$. At resolution $I = 6$, all 64 scaling coefficients are nonzero. On a cell where $f = c$, their normalization is

$$a_6[n] = \int f(t)\, 8\mathbf{1}_{Q_{6,n}}(t)\, \mathrm{d}t = c/8.$$

The Haar recursion is $a_{i-1}[n] = (a_i[2n] + a_i[2n+1])/\sqrt{2}$ and $d_{i-1}[n] = (a_i[2n] - a_i[2n+1])/\sqrt{2}$. Applying it gives

sequence	nonzero indices	values
a_0	0	2
d_0	0	$-5/8$
d_1	$0, 1$	$3\sqrt{2}/16$ at each index
d_2	$0, 3$	$1/8$ at each index
d_3	$1, 6$	$\sqrt{2}/8$ at each index
d_4, d_5	none	0

For example, $d_3[1] = 2^{3/2}(2/16 - 1/16) = \sqrt{2}/8$.

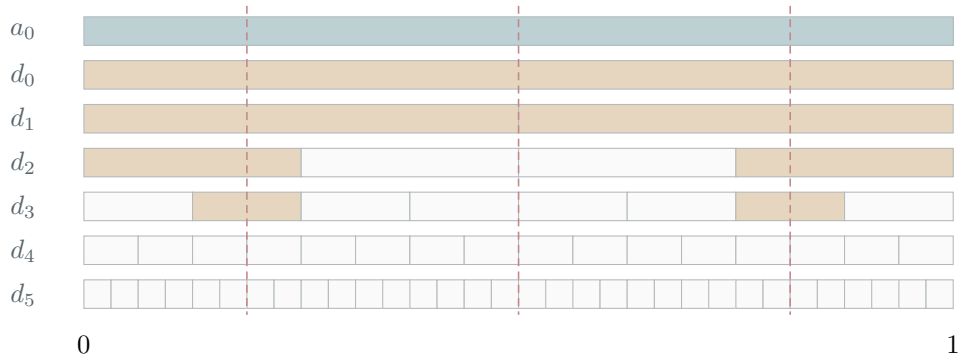


Figure 12.8: Filled cells mark nonzero coefficients, with rows ordered from coarse to fine. Dashed lines mark the three jumps. The finest two rows are entirely zero; this is a coefficient-location display, not a frequency-axis plot.

Only eight of the 64 DWT coefficients are nonzero, including the mean. No information was discarded: $f \in V_4$, so all details at $i \geq 4$ vanish exactly. Parseval checks the values:

$$\|f\|_{L^2}^2 = \frac{37}{8} = 2^2 + \left(\frac{5}{8}\right)^2 + 2\left(\frac{3\sqrt{2}}{16}\right)^2 + 2\left(\frac{1}{8}\right)^2 + 2\left(\frac{\sqrt{2}}{8}\right)^2.$$

12.10 Vanishing moments cancel polynomial behavior

Haar cancels constants, but a smoothly varying piece is not constant. A local polynomial is a better model. This motivates wavelets that cancel not only the mean but also higher-order moments.

Definition 12.1 (Vanishing moments)**.** A compactly supported wavelet $\psi \in L^2(\mathbb{R})$ has *at least p vanishing moments* if

$$\int_{\mathbb{R}} t^m \psi(t)\,\mathrm{d}t = 0, \qquad m = 0, 1, \ldots, p-1. \tag{12.9}$$

We call this a wavelet *having p vanishing moments*, with "p" meaning *at least* p unless stated otherwise. Compact support ensures these integrals exist; for noncompact wavelets, assume the corresponding weighted integrability.

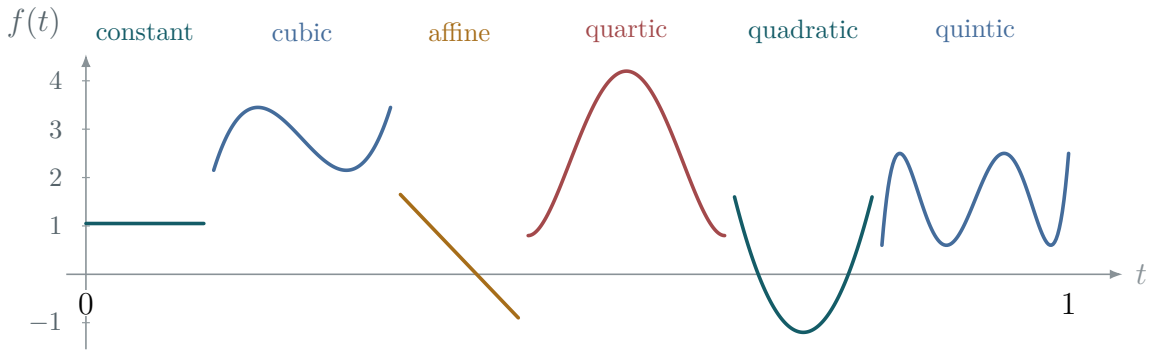


Figure 12.9: Six pieces, of degrees $0, 3, 1, 4, 2, 5$ from left to right. A wavelet with six vanishing moments cancels each piece whenever its entire support lies inside it. Gaps mark transitions; no global smoothness is assumed.

Proposition 12.2 (Local polynomial cancellation)**.** Let ψ have p vanishing moments. If f agrees with a polynomial of degree less than p on the support of $\psi_{i,n}$, then $\langle f, \psi_{i,n} \rangle_{L^2} = 0$.

Proof. It suffices to treat $f(t) = t^r$ on that support, with $r < p$. Set $u = 2^i t - n$. Conjugation is in the second argument, so

$$\begin{aligned} \int t^r \psi_{i,n}(t)^*\,\mathrm{d}t &= 2^{-i(r+1/2)} \int (u+n)^r \psi(u)^*\,\mathrm{d}u \\ &= 2^{-i(r+1/2)} \sum_{m=0}^{r} \binom{r}{m} n^{r-m} \left(\int u^m \psi(u)\,\mathrm{d}u \right)^* = 0. \end{aligned}$$

Linearity gives the claim. Only the compact wavelet support is involved; the polynomial need not belong to $L^2(\mathbb{R})$. □

12.11 More vanishing moments require a longer filter

This bound concerns *real, compactly supported, orthonormal dyadic MRA wavelets*, not arbitrary functions with vanishing moments or all biorthogonal constructions.

Theorem 12.3 (Minimum filter length)**.** Let a real FIR low-pass filter g_0 be supported in $\{0, \dots, L-1\}$, with both endpoint taps nonzero. Suppose it generates an orthonormal MRA, has $G_0(1) = \sqrt{2}$, and its wavelet has at least $p \geq 1$ vanishing moments. Then

$$\boxed{L \geq 2p.} \tag{12.10}$$

Here L includes internal zero taps. Unit energy and $G_0(1) = \sqrt{2}$ rule out $L = 1$. The last index $N = L - 1$ is odd: an even $N > 0$ would give the forbidden correlation $g_0[N]g_0[0] \neq 0$ at lag N.

Proof. Set $N = L - 1$. For $0 \leq m < p$, differentiating the Fourier integral gives $\widehat{\psi}^{(m)}(0) = (-\mathrm{j})^m \int t^m \psi(t)\, \mathrm{d}t = 0$. The two-scale and conjugate-mirror identities give

$$\widehat{\psi}(2\omega) = -\frac{\mathrm{e}^{-\mathrm{j}N\omega}}{\sqrt{2}} \widehat{g}_0(\omega + \pi)^* \widehat{\phi}(\omega).$$

Since $\widehat{\phi}(0) = 1$, its factor is nonzero near zero. Thus the moments vanish precisely when $\widehat{g}_0$ has a zero of order at least p at π, or $G_0(z) = (1 + z^{-1})^p Q(z^{-1})$ for a real polynomial Q. The substitution $z^{-1} = \mathrm{e}^{-\mathrm{j}\omega}$ preserves the zero's order because its derivative is nonzero.

The squared magnitude of a real FIR filter is an even trigonometric polynomial. Pairing opposite lags gives cosines, and the recurrence $\cos((r+1)\omega) = 2\cos\omega\cos(r\omega) - \cos((r-1)\omega)$ expresses each cosine as a polynomial in $\cos\omega$. Hence

$$\tfrac{1}{2}|\widehat{g}_0(\omega)|^2 = R(\cos\omega), \qquad \deg R \leq N.$$

Power complementarity gives $R(x) + R(-x) = 1$. The factorization of G_0 implies that $R(x)$ contains $(1 + x)^p$, because

$$|1 + \mathrm{e}^{-\mathrm{j}\omega}|^{2p} = 2^p (1 + \cos\omega)^p.$$

Consequently $1 - R(x) = R(-x)$ contains $(1 - x)^p$. The polynomial R' therefore has zeros of multiplicity at least $p - 1$ at both endpoints. Since $R(-1) = 0$ and $R(1) = 1$, it is not identically zero, so

$$N - 1 \geq \deg R' \geq 2(p - 1), \qquad L = N + 1 \geq 2p.$$

□

Why this also bounds the wavelet support. We measure support length as the length of the smallest interval containing the essential support, at scale zero. Keep the normalization $g_0[0]g_0[N] \neq 0$ and let $[a, b]$ be that interval for the compactly supported scaling function.

In

$$\phi(t) = \sqrt{2} \sum_{n=0}^{N} g_0[n]\phi(2t - n),$$

the leftmost possible point on the right is $a/2$, contributed by the $n = 0$ term. In a sufficiently small interval immediately to its right, all other terms vanish: their left endpoints are at least $(a+1)/2$. The nonzero endpoint tap cannot be canceled there. Thus the left endpoint of the right-hand side really is $a/2$. Equality of the two sides gives $a = a/2$, hence $a = 0$.

The same reasoning at the right edge gives $b = (b + N)/2$, hence $b = N$. This argument uses essential endpoints, so it does not require a nonzero value at a single endpoint.

For the conjugate-mirror filter $g_1[n] = (-1)^n g_0[N - n]$, both endpoint taps are nonzero. Since

$$\psi(t) = \sqrt{2} \sum_{n=0}^{N} g_1[n]\phi(2t - n),$$

the same uncancelled edge terms put the wavelet's support endpoints at 0 and N. Therefore

$$\boxed{\text{support length of } \phi = \text{support length of } \psi = L - 1 \geq 2p - 1.} \tag{12.11}$$

Other sign or translation conventions can move the support interval without changing its length. At scale i, its length is $2^{-i}(L - 1)$.

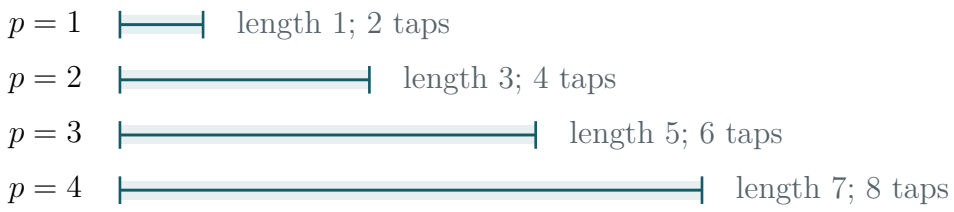


Figure 12.10: Minimum support spans at scale zero for the real orthonormal construction. More cancellation requires a longer footprint; dilation rescales every displayed length.

The Daubechies construction (1988) attains both bounds: $2p$ taps and support length $2p - 1$. The next two pages prove attainability.

12.12 Why the minimum length can be attained

We first build the *power response*, then recover a real filter from it. This follows the product-filter viewpoint: orthogonality is easiest to enforce on a nonnegative squared magnitude.

For $p \geq 1$, define

$$R_p(x) = c_p \int_{-1}^{x} (1-u^2)^{p-1}\,\mathrm{d}u, \qquad c_p = \left[\int_{-1}^{1} (1-u^2)^{p-1}\,\mathrm{d}u\right]^{-1}.$$

The integrand is an even polynomial of degree $2p-2$. Therefore R_p is a polynomial of degree $2p-1$ satisfying

$$R_p(-1) = 0, \qquad R_p(1) = 1, \qquad R_p(x) + R_p(-x) = 1.$$

The last identity follows by substituting $u \mapsto -u$ in the integral. Near $x = -1$, the integrand has a zero of order $p-1$, so its integral has a zero of exactly order p. Thus R_p has precisely the endpoint flatness needed in the preceding proof.

Set

$$F_p(\omega) = 2R_p(\cos\omega). \tag{12.12}$$

Then $F_p \geq 0$, $F_p(0) = 2$, and

$$F_p(\omega) + F_p(\omega + \pi) = 2.$$

Also, F_p vanishes to order $2p$ at $\omega = \pi$, since $1 + \cos\omega$ vanishes to order two there. It is strictly positive everywhere else in a period.

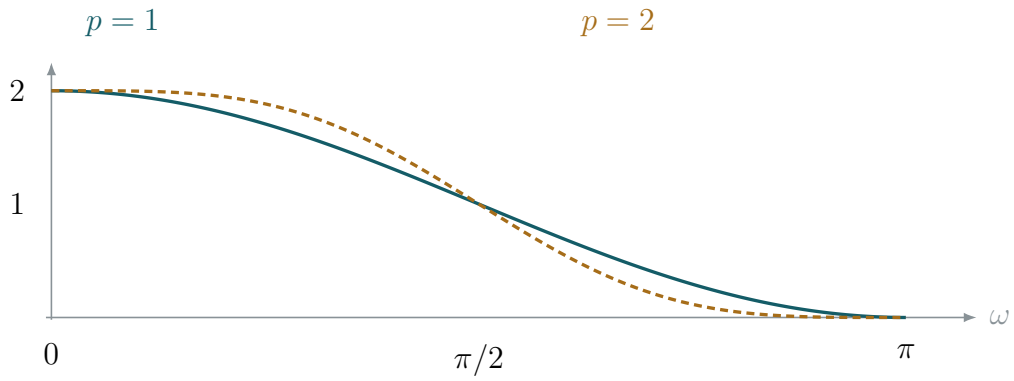


Figure 12.11: The first two minimum-length power responses. The two-moment design has flatter behavior at zero and π than Haar. The curves cross at the common half-band value one.

For $p = 1$, $R_1(x) = (1+x)/2$, recovering the Haar power. For $p = 2$, integration gives $R_2(x) = 1/2 + 3x/4 - x^3/4$. The flatter endpoints improve cancellation, at the cost of a longer filter.

Recovering a real filter from its power. Put $N = 2p - 1$ and form the real Laurent polynomial

$$\mathcal{P}(z) = 2R_p\left(\frac{z + z^{-1}}{2}\right).$$

On the unit circle, $\mathcal{P}(\mathrm{e}^{\mathrm{j}\omega}) = F_p(\omega)$. We seek a real filter satisfying $\mathcal{P}(z) = G_0(z)G_0(z^{-1})$.

The ordinary polynomial $z^N\mathcal{P}(z)$ has degree $2N$, a nonzero constant term, and reciprocal real coefficients. Its roots therefore occur in reciprocal and conjugate pairs: if r is a root, then so are $1/r$, r^*, and $1/r^*$, with multiplicity. Its only unit-circle root is -1, of multiplicity $2p$. This follows from the positivity and zero order of F_p established above.

Choose p copies of the root -1. From the remaining reciprocal pairs, choose the roots inside the unit circle. There are $N - p = p - 1$ of them, counted with multiplicity; this set is closed under complex conjugation. The resulting degree-N polynomial

$$Q(z) = \prod_{r \text{ chosen}} (z - r)$$

has real coefficients. Both $z^{-N}Q(z)$ and its reciprocal partner together have exactly the zeros of $\mathcal{P}$, with the same multiplicities. Hence

$$Q(z)Q(z^{-1}) = c\,\mathcal{P}(z)$$

for a real constant c. At $z = 1$, both sides are positive, so $c > 0$. Rescaling Q and choosing its overall sign gives

$$G_0(z) = z^{-N}Q(z), \qquad G_0(1) = \sqrt{2}, \qquad G_0(z)G_0(z^{-1}) = \mathcal{P}(z).$$

This is a real FIR filter of length $N + 1 = 2p$, with a zero of order p at $z = -1$ and the required power complementarity.

Finally, for $|\omega| \le \pi/2$, $\cos\omega \ge 0$, so $|\hat{g}_0(\omega)|^2 = 2R_p(\cos\omega) \ge 1$. The sufficient scaling-function theorem from Lecture 11 applies: the normalized infinite product yields an orthonormal MRA. Its conjugate-mirror wavelet has p vanishing moments and support length $2p-1$. This completes the attainment argument. Selecting the inside roots is the usual minimum-phase choice, apart from the required unit-circle zeros.

12.13 The four-tap, two-moment example

The first case beyond Haar has $p = 2$ and $L = 4$. Write the real low-pass taps as (a, b, c, d). Normalization and the zero at π give

$$a + b + c + d = \sqrt{2}, \qquad a - b + c - d = 0.$$

The second vanishing moment imposes $-b + 2c - 3d = 0$. The only nontrivial even-shift correlation is $ac + bd = 0$.

Put $u = 1/\sqrt{2}$. The first two equations give $a + c = b + d = u$. Substitution into the third gives $a + d = u/2$, hence

$$c = u - a, \qquad d = u/2 - a, \qquad b = u/2 + a.$$

The correlation condition becomes

$$a(u - a) + (u/2 + a)(u/2 - a) = 0 \quad \Longleftrightarrow \quad 2a^2 - ua - u^2/4 = 0.$$

Taking $a = u(1 + \sqrt{3})/4$ produces

$$\boxed{g_0 = \frac{1}{4\sqrt{2}}(1 + \sqrt{3},\ 3 + \sqrt{3},\ 3 - \sqrt{3},\ 1 - \sqrt{3}).} \tag{12.13}$$

The other root reverses the taps. We have not separately imposed unit norm: it follows from

$$a^2 + b^2 + c^2 + d^2 = (a + c)^2 + (b + d)^2 - 2(ac + bd) = 1.$$

All other nonzero even shifts have disjoint support. Thus the even translates are orthonormal. Its factorization is

$$G_0(z) = \frac{(1 + z^{-1})^2}{4\sqrt{2}}\big[(1 + \sqrt{3}) + (1 - \sqrt{3})z^{-1}\big].$$

The double zero is explicit. Moreover, its squared magnitude is $1 + \frac{3}{2}\cos\omega - \frac{1}{2}\cos^3\omega = 2R_2(\cos\omega)$, so the scaling-function existence condition checked above holds.

The high-pass filter is $g_1 = (d, -c, b, -a)$, and the analysis filters are $h_\ell[n] = g_\ell^*[-n]$. This wavelet cancels affine pieces, unlike Haar, and has support length three rather than one.

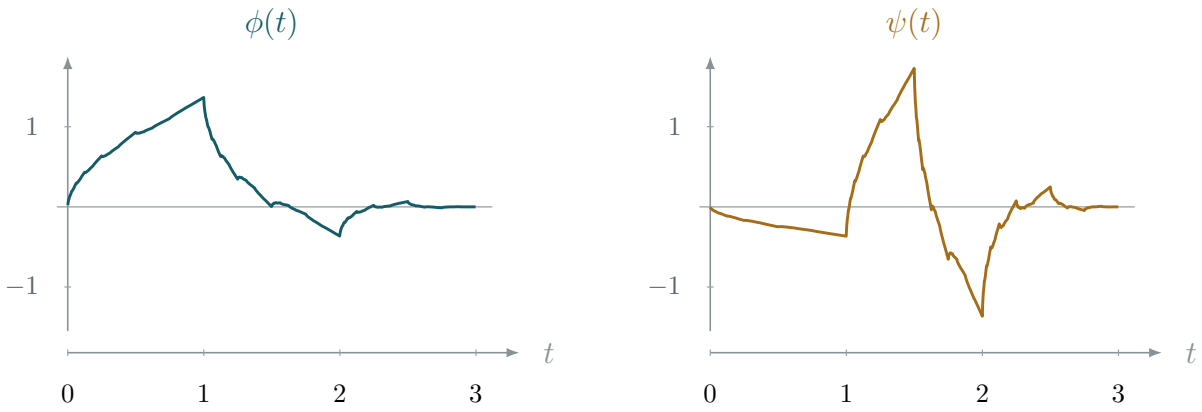


Figure 12.12: The four-tap scaling function and conjugate-mirror wavelet on a common vertical scale. The curves are numerical approximations from ten box-seeded low-pass refinements and one high-pass refinement. Both support spans are $[0, 3]$; the wavelet has two vanishing moments.

The design tradeoff. More vanishing moments improve cancellation on smooth pieces, but increase the minimum support and filter length. A longer support overlaps more breakpoints. If $\operatorname{supp} \psi \subseteq [0, L_\psi]$, the condition that $\psi_{i,n}$ straddle a breakpoint τ is

$$2^i\tau - L_\psi < n < 2^i\tau.$$

This interval contains at most $\lceil L_\psi \rceil$ integer indices. Thus S breakpoints can affect at most $S\lceil L_\psi \rceil$ wavelets at each scale. For piecewise polynomials of degree below p, every other detail coefficient vanishes by local polynomial cancellation.

Moments p	Minimum filter length	Minimum support length
1	2	1
2	4	3
3	6	5
4	8	7

Daubechies wavelets attain these bounds in the real orthonormal MRA setting. Symmlets use other phase choices to improve symmetry while retaining compact support and vanishing moments. Orthogonalized spline wavelets need not be compactly supported; biorthogonal spline systems are outside the scope of this bound.

Octave bands, localization, and polynomial cancellation explain wavelet sparsity even when every sample is nonzero. A global sinusoid can instead be sparse in Fourier coordinates: the representation should match the signal.